a sine wave. Because the new signal's sole harmonics are at the switching frequency, these can be easily filtered away. This is a good approach, but it adds to the design complexity.

Input inrush current/Surge current. When you turn on a power source or electrical equipment, it generates a temporary current called inrush current. This is due to the high initial current needed to charge the capacitors, inductors, and transformers. The inrush current is also known as the switch-on surge or input surge current. It is an important performance indicator for switching power supplies.

In the case of a switched-mode power supply (SMPS), the output capacitor, which is normally big in value, can hold a huge amount of charge that is comparable to a short-circuit when not energised. The switches in the SMPS link the input source to the empty capacitor when an energy source is attached to the SMPS. This creates a low impedance path for the inrush current to flow to the capacitor. The voltage across the capacitor rises, but slowly decreases thereafter. That being said, the inrush current is large enough to damage the switches in the supply circuit, and hence needs to be designed as small as possible.

To control the inrush current, a 'pre-charging' circuit may be used. The conventional design is to connect a thermistor (NTC) in series on the live wire at the input of the switching power supply. For switching power supply with a higher power, a relay is connected in parallel to the thermistor (NTC) at the same time to reduce device loss and improve reliability when the product works normally.

Let us now focus on analysing the causes of contact short-circuit failure after incorporating the relay.

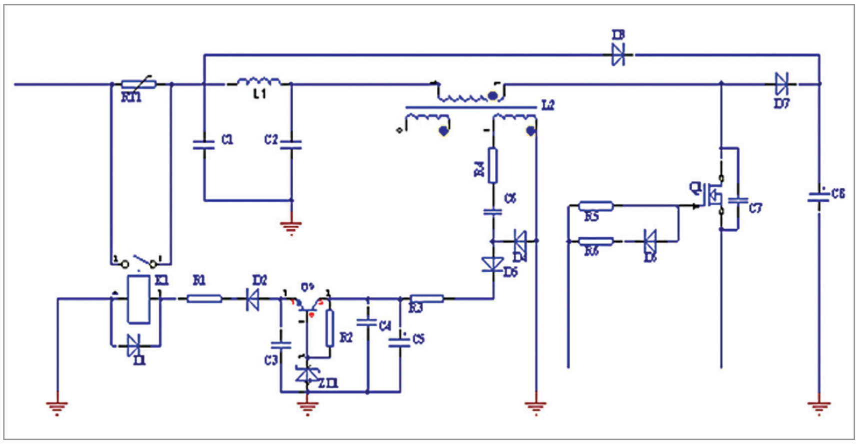

Fig. 2: PFC circuit

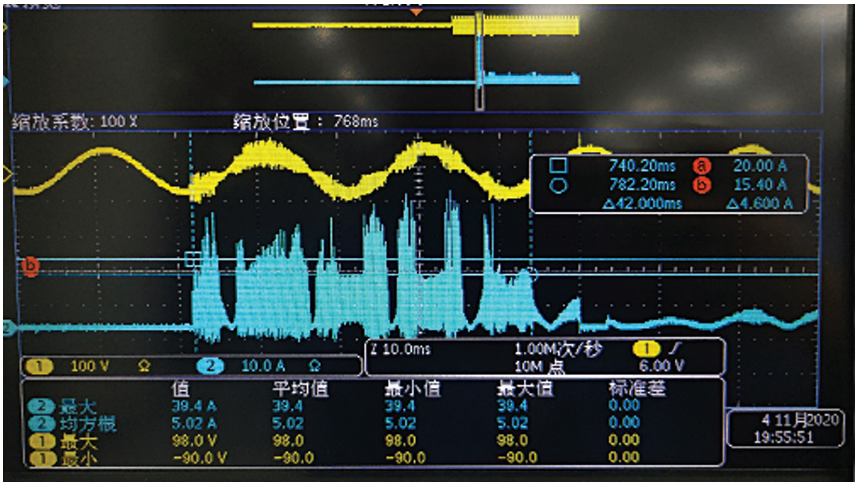

Fig. 3: Conduction waveform of relay contact (the yellow waveform is the input voltage and the blue one is the relay contact current)

Finding the fault

When the PFC circuit in Fig. 2 was applied to the product, it was found that the inrush current exceeded the standard value. This result was completely different from the theoretical calculation. According to Fig. 2, the devices that affect the input inrush current are mainly the thermistor RT1 and relay K1.

After testing the thermistor and relay it was observed that the thermistor resistance is normal, and the normally open contact of the relay is taken in (sucked) without a power supply. This means that the relay is the abnormal device here.

When the relay is not powered on and the normally open contact has been closed, the relay contact is short-circuited. There are generally three possibilities accounting for this failure.

Case 1. The operation frequency of the relay is too high, and the times have exceeded the switching times that the relay can bear

Case 2. The surrounding temperature of the relay is too high

Case 3. The surge current flowing through the relay is too large

After analysing each of these three cases, it was noticed that the failure was not caused by the high operation frequency of the relay or the high surrounding temperature. Hence, cases 1 and 2 could be ruled out. The failure was caused by the large surge current flowing through the relay.